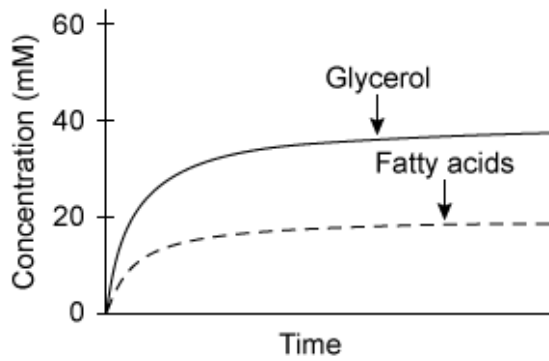


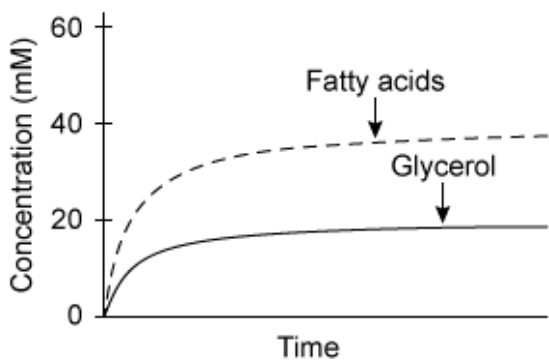
Which of the following graphs best depicts the glycerol and fatty acid product levels as a 20 mM solution of a glycerophospholipid undergoes hydrolysis of its ester and phosphoester bonds?

☐ A.



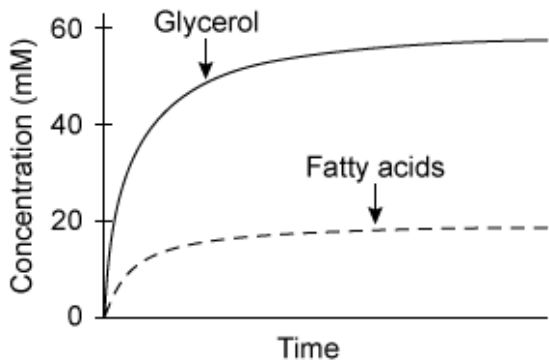
[7%]

☒ B.



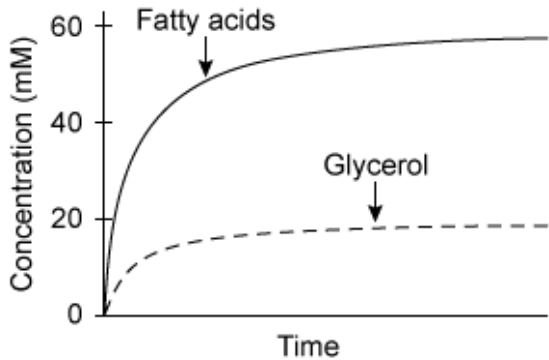
[38%]

☐ C.



[6%]

☐ D.



[46%]

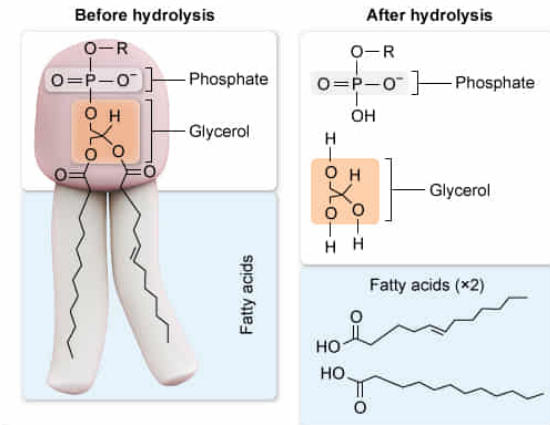
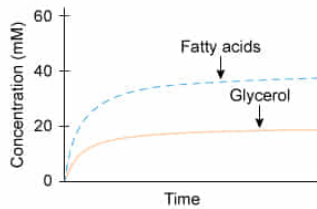
Correct

38% answered correctly

Explanation:

Glycerophospholipid structure and hydrolysis products

Hydrolysis of a glycerophospholipid produces two fatty acids for every glycerol molecule produced



Phospholipids are major components of cell and organelle membranes. The **most common type of phospholipid** in a membrane is the **glycerophospholipid** (as opposed to **phosphosphingolipids**). Glycerophospholipids consist of a **glycerol backbone** bound to a **phosphate group** through a **phosphoester bond** and to **two fatty acids** through **ester bonds**. The phosphate group may also be linked to other groups such as choline, serine, or ethanolamine to form a phosphate headgroup.

Each of the bonds to the glycerol backbone (ie, ester, phosphoester) can undergo **hydrolysis**, which breaks the bond as water is added. When all bonds are broken, **glycerol is released** along with the **two fatty acids** and the **phosphate headgroup**. The initial concentration of glycerophospholipids was 20 mM. Because one glycerol molecule is released per glycerophospholipid, the final concentration of glycerol approaches 20 mM as the reaction proceeds.

Because two fatty acids are initially attached to each glycerol molecule, the **final concentration of fatty acids** (40 mM) is **twice that of glycerol**. Only the graph shown in Choice B depicts the correct final concentrations of glycerol and fatty acids.

(Choice A) This graph shows twice the final amount of glycerol (40 mM) as fatty acids (20 mM). These levels should be reversed.

(Choice C) This graph shows a final glycerol level of 60 mM, which is three times larger than the 20 mM level shown for fatty acids. Neither final concentration is correct.

(Choice D) This graph shows a final fatty acid concentration three times greater than the glycerol concentration. This result would be observed in the hydrolysis of **triglycerides**, which have three fatty acids attached to a glycerol backbone, but not for glycerophospholipids.

Educational objective:

Glycerophospholipids consist of a glycerol backbone linked to a phosphate group and two fatty acids. The phosphate may also be linked to additional groups.

Time spent:0 sec**Subject:**

Biochemistry

QID:403706**System:**

Carbohydrates, Nucleotides, and Lipids